Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports

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Abstract. Operational flood streams contain duplicate, incomplete, and adversarial reports, while clustering and priority scores are often evaluated without measuring their scheduling consequences. We audit a fail-closed batch pipeline in which observable reports form a sparse graph, unresolved reports go to review, duplicate families feed a bounded score, and evaluator-only truth is joined after predicted-cluster scheduling. The clustering benchmark contains 80 synthetic runs from generator 3.0.0: additive Louvain has mean ARI 0.9191, product Leiden 0.9165, and product Louvain 0.9072. Product minus independently tuned additive is -0.0118 (bootstrap 95% CI $[-0.0226, -0.0024]$), but its Holm-adjusted test is not significant ($p = .2560$). A matched-density comparison reverses the point estimate but is also inconclusive. In a separate candidate-4.1 bundle, the revised score has NDCG@5 0.6669 versus 0.6763 for a random comparator and does not show a general alignment advantage; exact duplicates are invariant but coordinated high-confidence campaigns remain a failure mode. Dispatch results similarly show no general benefit over the legacy score and a clear cost relative to nearest-first. These are synthetic, within-generator diagnostics: boundedness and clustering accuracy do not establish source authenticity, policy validity, field benefit, or deployment readiness.

Keywords: Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

1 Introduction and Related Work

During emergencies, crisis streams mix repeated reports of one event with nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency-message research addresses volume, verification, and event extraction [9]. CrisisMMD and FloodNet capture useful disaster modalities, although FloodNet is an aerial post-flood scene-understanding dataset rather than a multimodal report stream [1,8]. These resources do not jointly provide physical-incident linkage, rescue priority, and field dispatch outcomes.

Social sensing and event extraction. Social-media users can act as real-time sensors [9], but the usual endpoint is event detection or classification rather than the consequences of sending a field resource to a predicted incident.

Consolidation and clustering. Graph community detection with Louvain or Leiden [3,13], and density methods such as DBSCAN and HDBSCAN [4] offer plausible consolidation mechanisms. Multiplicative spatial–range terms occur in bilateral filtering [12]; we adapt that structure to a product graph, while testing the exact gate and its bounds here.

Humanitarian logistics and dispatch. Relief-routing models expose competing equity, efficacy, and efficiency objectives [5]. We use this perspective to motivate a bounded priority ranking, but boundedness is not an expert-endorsed rescue policy. A key gap is propagation of false splits, merges, and noise into a scheduler that cannot consult incident truth.

We address these gaps with a fail-closed batch pipeline: reports with observable location and time enter a sparse graph, incomplete reports go to manual review, duplicate families feed a bounded ranking, and evaluator-only truth joins after predicted-cluster scheduling. We ask: (RQ1) does independently selected product composition improve clustering over additive composition? (RQ2) does the duplicate-aware score align with an independently generated benefit under report-level stress tests? (RQ3) how do predicted split, merge, and noise errors propagate into dispatch? The questions use two explicitly separated artifact bundles.

The contributions are (i) an observable-only pipeline with explicit review and evaluator boundaries; (ii) auditable threshold statements and a proposed duplicate-aware score; and (iii) a paired diagnostic reporting clustering, ranking, dispatch, and adverse results. The evidence remains synthetic and within-generator, so the artifact is not a validated rescue policy.

2 Observable Pipeline and Methods

Figure 1 fixes the information boundary. Receipt-visible fields flow through graph construction, ranking, and a batch scheduler. Missing location or event time routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

2.1 Report Attributes and Missingness

Each report is represented by the compact tuple

$$r_i = (L_i, T_i, F_i, E_i, N_i, V_i, Q_i),$$

where L_i is a GPS location and T_i is the observable report timestamp ("created_at"). Flood evidence F_i and urgency evidence E_i are in $[0, 1]$; $N_i \in \mathbb{Z}_{\geq 0}$ is the reported trapped-person count; $V_i \geq 0$ is a vulnerability-evidence value; and $Q_i \in [0, 1]$ is a provenance/confidence heuristic. Q_i is not a calibrated probability of truth. The implementation also carries observable support fields such as image presence, source type, province, and missing-field flags; these support Q_i and duplicate fingerprints but are not additional latent labels.

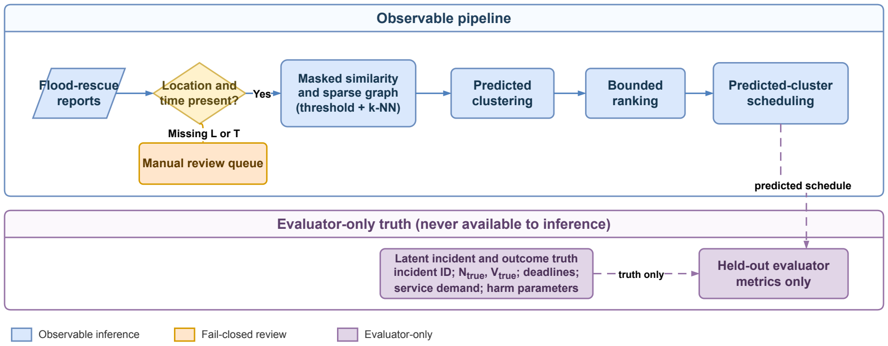

Fig. 1. Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

The score Q_i is computed from observable image presence and a capped count n_i^{corrob} of nearby corroborating payloads using the proposed heuristic

$$\begin{aligned} Q_i &= \operatorname{sigmoid}\bigl(b_0 + b_1 \mathbf{1}\{\text{image}_i\} + b_2 \log(1 + n_i^{\text{corrob}})\bigr)\,, \\ &(b_0, b_1, b_2) = (-.2, 1.4, .9). \end{aligned}$$

Here $\operatorname{sigmoid}(x) = (1 + e^{-x})^{-1}$. This is a pipeline rule, not a calibrated truth model. No independent receipt timestamp or source family is assumed.

Following the general missing-variable kernel problem [10], we use the following fail-closed construction. The admission indicator $a_i = \mathbf{1}\{L_i, T_i \text{ are observed}\}$. Thus $a_i a_j = 0$ prevents an edge involving a report whose geographic or temporal separation is undefined; such a report follows the manual-review path. Missing F_i or E_i is not zero-imputed: let $I_F(i,j), I_E(i,j)$ indicate that both reports share the corresponding observation and let $m_{ij} = I_F(i,j) + I_E(i,j)$; then

$$C_{ij} = \frac{m_{ij}}{2} \exp\left[-I_F(i,j)\frac{|F_i - F_j|}{\tau_F} - I_E(i,j)\frac{|E_i - E_j|}{\tau_E}\right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0.$$

When $m_{ij} = 2$, both contextual fields are used; when $m_{ij} = 1$, the maximum contextual similarity is discounted to $1/2$; and when $m_{ij} = 0$, no contextual agreement is inferred. In the current 80-run artifact, every report has location, event time, flood, and urgency. Hence $a_i = 1$ and $m_{ij} = 2$ throughout the evaluated data, reducing the expression to

$$C_{ij} = \exp\left[-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E}\right].$$

The missingness extension is specified and unit-tested, but missingness robustness is not claimed as an empirical result of this benchmark.

2.2 Similarity Graph and Derived Bounds

For Haversine distance d_{ij} and time difference Δt_{ij}, define the Gaussian affinity [7] and exponential temporal decay [15]:

$$\begin{aligned} G_{ij} &= \exp[-d_{ij}^2/(2\sigma^2)], & T_{ij} &= \exp[-\Delta t_{ij}/\tau_t], \\ w_{ij}^{\times} &= a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij}), & w_{ij}^{+} &= a_i a_j(\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \end{aligned} \tag{1}$$

The exponential form is a precedent, not a claim that the complete missingness-aware expression is copied from prior work [2]. The product is a spatial gate motivated by spatial–range products such as bilateral filtering [12]; both compositions and all weights here are defined by this paper. All scales $\sigma, \tau_t, \tau_F, \tau_E$ are strictly positive and the weights are nonnegative. The implementation first forms a common spatial candidate pool of at most $\max(64, 4k)$ neighbours per eligible report, including all boundary-distance ties before canonical truncation. It computes an empirical nonzero-weight quantile, retains strict above-threshold edges, keeps the union of endpoint-wise top-k edges, and applies Louvain. Product and additive candidates have the same pool and grid, but each family is selected independently; the estimand is therefore a pipeline comparison, not an operator-only causal effect.

Theorem 1 (Product edge localization). *Let $B = \beta + \gamma > 0$. If a product edge with threshold $0 < \theta < B$ is retained, then $d_{ij} < \sigma\sqrt{2\log(B/\theta)}$. For $\theta \geq B$ the strict retained set is empty; for $\theta \leq 0$ the threshold alone gives no finite cutoff.*

Proof. Since $a_i a_j \leq 1$ and $T_{ij}, C_{ij} \leq 1$, $w_{ij}^{\times} \leq B\exp[-d_{ij}^2/(2\sigma^2)]$. Combining $\theta < w_{ij}^{\times}$ with positivity and taking logarithms gives the bound; the endpoint cases follow from the same inequality.

This is an edge statement. A component-diameter claim additionally needs a bounded observed hop diameter; long transitive chains remain possible.

Corollary 1 (Conditional component bound). *In the finite product region, a connected component with unweighted observed hop diameter $h < \infty$ and geographic diameter D satisfies $D < hr_\theta^{\times}$, where $r_\theta^{\times} = \sigma\sqrt{2\log(B/\theta)}$. A singleton has $h = D = 0$.*

The result follows from the triangle inequality along a path of at most h retained edges. It is a conditional post-hoc component bound, not an ex-ante compactness guarantee: the pipeline does not control h.

For the additive form, assume $\alpha > 0$. If $B < \theta < B + \alpha$, every retained edge satisfies

$$d_{ij} < r_\theta^{+} = \sigma\sqrt{2\log\left(\frac{\alpha}{\theta - B}\right)}.$$

If $\theta \geq B + \alpha$, the strict retained set is empty; if $\theta \leq B$, no non-trivial threshold-induced geographic cutoff exists in general. Indeed, retention gives $\theta < \alpha G_{ij} +$

$\beta T_{ij} + \gamma C_{ij} \leq \alpha G_{ij} + B$. Thus product composition has a finite geographic edge bound throughout its nonempty positive-threshold domain, whereas additive composition has such a bound only above the maximum non-geographic contribution. Neither statement alone implies better empirical clustering.

2.3 Duplicate-Aware Bounded Ranking

Exact duplicate payloads share a canonical fingerprint. Near duplicates are joined by deterministic connected components under the observable envelope; this transitive rule is not complete linkage. Let g index the resulting evidence families. The revised estimator uses confidence-gated family maxima and avoids treating overlapping reports as disjoint people. Write $\widetilde{N}_i = \min(N_i, N_{\mathrm{ref}})$ and $\widetilde{V}_i = \min(V_i, V_{\mathrm{cap}})$ for the nonnegative clipped claims:

$$\bar{E} = \frac{\sum_g \max_{i \in g} Q_i E_i}{\max(1, \sum_g \max_{i \in g} Q_i)}, \qquad \bar{F} = \max_{g, i \in g} Q_i F_i,$$

$$\bar{N} = \min\left(1, \frac{\log(1 + \max_{g, i \in g} Q_i \widetilde{N}_i)}{\log(1 + N_{\mathrm{ref}})}\right), \qquad \bar{V} = \max_{g, i \in g} Q_i \widetilde{V}_i,$$

$$P = \left[1 + (\mu - 1)\tanh(\bar{V}/s)\right](\omega_E \bar{E} + \omega_F \bar{F} + \omega_N \bar{N}). \tag{2}$$

The candidate source snapshot associated with the RQ2/RQ3 artifacts specifies $\omega = (.34, .33, .33)$, $\mu = 2$, $s = 10$, $N_{\mathrm{ref}} = 500$, and $V_{\mathrm{cap}} = 50$. The formula is a proposed policy heuristic: maxima limit duplicate multiplicity, the logarithm compresses heavy-tailed demand, clipping bounds individual evidence, and the hyperbolic tangent saturates vulnerability amplification. Reliability-aware truth discovery motivates using Q_i but does not derive these maxima [14]. Transformation, normalization, and weights require sensitivity and expert validation [11,5]. Comparators are a multiplicity-sensitive legacy score, urgency-only, population-only, a simple linear score, stable random, and—for dispatch—nearest-first.

3 Experimental Design

Data contract and splits. The artifact contains 80 independently seeded synthetic datasets geographically anchored to the Copernicus EMSR848 Central Viet Nam flood activation. Copernicus EMS, OpenStreetMap, and WorldPop provide geographic or historical anchors; rescue incidents, reports, duplicates, coordinated campaigns, vulnerability, and incident labels are synthetic. Every run has 16 latent incidents, 288–370 reports, exact and near duplicates, and three coordinated nonincident campaigns of 20 reports each. Across all runs are 26,288 reports: 17,901 genuine, 3,587 duplicate, and 4,800 coordinated nonincident reports. Runs 1–20 are reserved for development, runs 21–40 for calibration, and runs 41–80 for held-out testing. All runs use one fixed generator regime, so the experiment measures within-generator variation, not mechanism-shift OOD transfer. Inference reads only `algorithm_input.json`; ground truth is opened after predictions for a test run have been produced.

Table 1. RQ1 benchmark-test summary over 40 runs. Intervals are bootstrap 95% confidence intervals; diameter is descriptive.

Method	ARI [95% CI]	Pair F1	Mean diam. (km)
Additive Louvain	.9191 [.8995,.9381]	.9247	.3477
Product Leiden	.9165 [.8979,.9351]	.9223	.3547
Convex Louvain	.9135 [.8940,.9320]	.9195	.3547
Product Louvain	.9072 [.8864,.9271]	.9138	.3585
Matched-density Additive	.8973 [.8763,.9175]	.9046	.3743
Coordinate K-means	.8176 [.7976,.8384]	.8292	.2977
Spatial agglomerative	.6235 [.5964,.6494]	.6506	4.1648
HDBSCAN	.5205 [.4565,.5811]	.5787	1.2527
Geo-time DBSCAN	.4978 [.4589,.5366]	.5450	.6244
DBSCAN, all features	.4741 [.4273,.5201]	.5488	.9340

Selection and comparators. The current notebook evaluates complete small grids on the 20 calibration runs and selects mean ARI, breaking ties by standard deviation and canonical configuration JSON. Product and additive use fixed $\tau_F = .25$, $\tau_E = .35$, and $\beta = \gamma = .5$. Selected product Louvain uses $\sigma = 700$ m, $\tau_t = 60$ min, quantile .90, $k = 8$, and resolution 1.2. Selected additive Louvain uses $\alpha = 1$, $\sigma = 700$ m, $\tau_t = 60$ min, quantile .95, $k = 8$, and resolution 1.2. Because selected quantiles differ, this is an independently calibrated pipeline comparison, not a matched-density operator comparison. Comparators are a convex similarity, product Leiden, geo-time DBSCAN, DBSCAN/HDBSCAN on scaled observable features, spatially constrained agglomerative clustering, and coordinate K-means. The implemented geo-time DBSCAN lacks ST-DBSCAN's non-spatial criterion and is named accordingly.

Endpoints and inference. ARI on incident-linked reports [6] is the primary endpoint; NMI, pairwise F1, geographic diameter, runtime, false destinations, fake absorption, noise rejection, retained-edge fraction, and mean degree are descriptive or operational diagnostics. The benchmark includes a matched- density product–additive comparison. Seed-paired ARI differences use 10,000 bootstrap resamples, Wilcoxon signed-rank tests, and Holm correction. The priority stress tests and predicted-cluster dispatch evaluation use independent benefit, deadline, service, access-delay, and harm outputs reported in Section 4; no clustering metric is substituted for those outcomes.

4 Results

Additive Louvain has the highest mean ARI. Product minus independently tuned additive is -0.01184 with CI $[-0.02257, -0.00240]$, with 9 wins, 19 ties, and 12 losses; raw $p = .0987$ and Holm $p = .2560$. Product minus matched-density additive is $+0.00989$ with CI $[-0.00100, 0.02082]$ and Holm $p = .2560$. Product Leiden versus Product Louvain has Holm $p = .1120$. Thus neither composition advantage is established. Product and matched-density additive have nearly equal retained-edge fractions (.03011 and .03021) and mean degrees (9.90 and

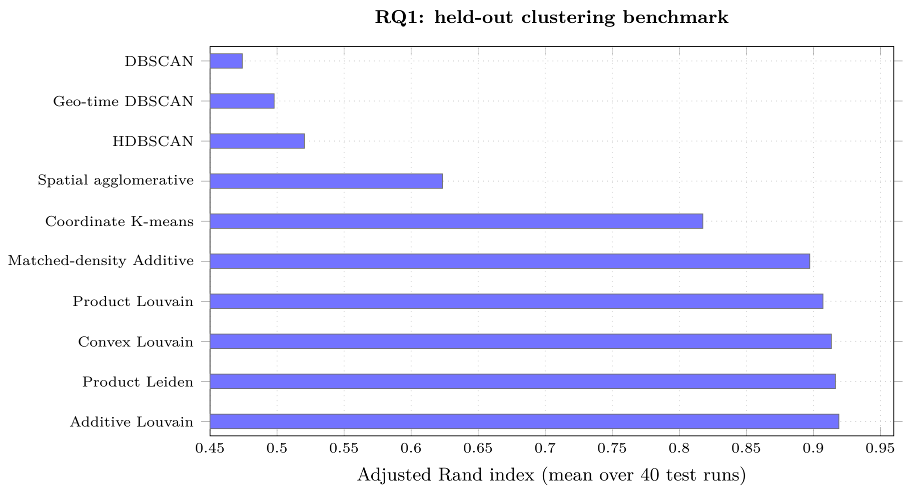

Fig. 2. RQ1 held-out ARI means generated from the benchmark summary CSV.

Table 2. Headline RQ2/RQ3 results from the candidate-4.1 artifacts. Positive paired effects favor the revised candidate; lower-is-better metrics use the artifact's oriented sign convention.

Question	Contrast and metric	Effect	Holm p
RQ2	Revised NDCG@5 (absolute)	.6669	–
	Random NDCG@5 (absolute)	.6763	–
	Revised − legacy NDCG@5	+.0121	.5882
	Exact-duplicate priority drift	0	–
	Coordinated-campaign drift	.1688	–
RQ3	Revised − legacy, Product harm	−14.34	.675
	Revised − legacy, Product deadline	+.0188	.160
	Revised − nearest, Product harm	−69.88	1.8×10^{-7}
	Revised − nearest, Product deadline	−.1771	3.0×10^{-15}
	Product − matched Additive harm	−18.34	.000151

9.94), so the matched-density contrast is a useful diagnostic but not a causal operator comparison.

The product pipeline has zero fake absorption and zero fake rejection but a 16.49% false-destination rate; fake reports avoid genuine destinations while still forming fake-only destinations. This is an adverse operational result, not evidence of misinformation robustness.

For RQ2, revised NDCG@5 is .6669 (CI [.6306,.7018]), Spearman is -.0304 (CI [-.1165,.0549]), and top-five overlap is .285 (CI [.2349,.3400]). The revised score is above legacy on NDCG@5 by .0121, but the Holm-adjusted contrast is not significant; it is below population-only (.6709) and random (.6763). Exact duplication at 2×, 5×, and 10× produces zero revised priority drift. A coordinated high-confidence campaign produces drift .1688 and false-priority lift .8828;

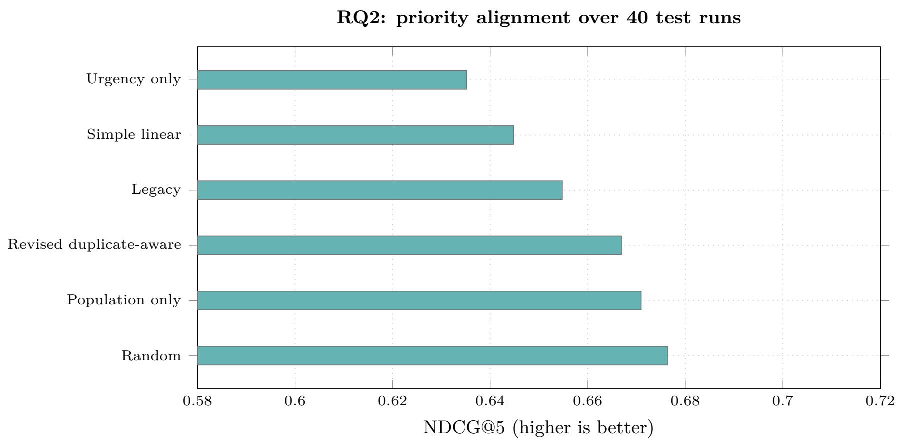

Fig. 3. RQ2 priority-alignment comparison generated from the candidate-4.1 summary CSV.

revised drift is worse than legacy by $-.1379$ (Holm $p \approx 9.1 \times 10^{-12}$). Boundedness therefore does not guarantee ranking reliability.

For RQ3, revised versus legacy on the Product partition has harm effect -14.34 (Holm $p = .675$) and deadline effect $+.0188$ (Holm $p = .160$), so no general improvement is established. Revised is significantly worse than nearest-first on both harm and deadline. Product is also worse than matched- density Additive under the revised policy on harm, and worse than oracle on the corresponding dispatch outcomes. Predicted split, merge, and fake-only destinations therefore propagate into operational consequences.

5 Discussion and Threats to Validity

The results separate mathematical safeguards from empirical utility. Product gating supplies a finite edge bound in its positive-threshold domain, but the independently tuned product pipeline does not outperform additive clustering. The matched-density point estimate reverses the direction, yet its adjusted test remains inconclusive. The graph theorem is therefore a localization property, not evidence of a preferred composition.

The revised score removes exact-duplicate inflation and limits multiplicity, but it does not show a general alignment advantage. Its coordinated-campaign failure is especially important: an adversary can create a high-confidence pattern that remains numerically bounded while still perturbing the ranking. Dispatch results confirm that a bounded score cannot substitute for a validated operational objective; clustering errors and fake-only destinations are transmitted to scheduling.

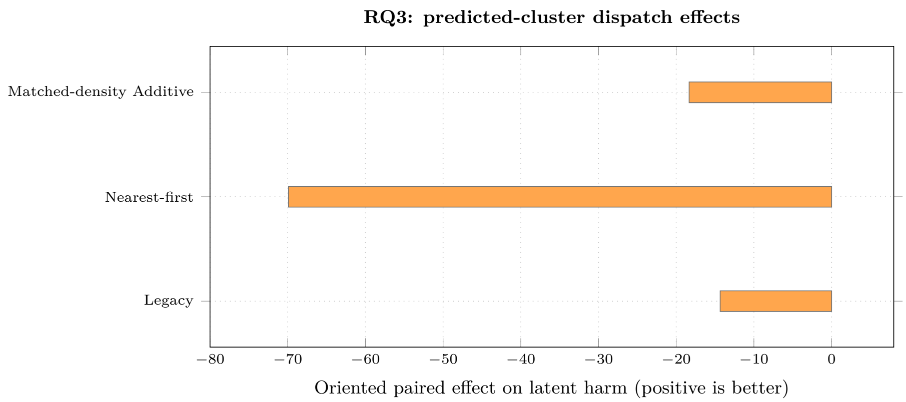

Fig. 4. RQ3 Product-partition paired effects on latent harm generated from the dispatch comparison CSV; positive values are improvement-oriented.

All reports, incidents, relevance signals, and dispatch outcomes are synthetic. RQ1 varies seeds within generator 3.0.0, while RQ2/RQ3 use a separate candidate-4.1 bundle; neither tests mechanism-shift OOD transfer or field performance. RQ1 does not provide an empirical missingness test because all evaluated reports contain the four core attributes. No expert panel validated the priority weights, caps, or utility. The study also does not measure extraction accuracy, device latency, memory, energy, provenance authenticity, or real-world harm. These limitations preclude policy or deployment claims.

6 Artifact and Conclusion

The accompanying artifact contains the 80-run dataset, benchmark notebooks, calibration grid, selected configurations, per-run test results, paired comparisons, and the deterministic figure-generation script. The publication figures are generated from the result CSVs; the original artifact PNGs remain available for audit. Every reported number is generated from an accepted CSV row and its documented generator and configuration metadata.

In conclusion, the study establishes conditional graph bounds and documents competitive within-generator clustering, exact-duplicate invariance, and important negative results for ranking and dispatch. It does not establish ranking alignment, dispatch benefit, misinformation robustness, policy validity, or deployment readiness. The contribution is an auditable stress-test boundary around those claims, not a validated rescue policy.

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